

Real Estate Price Prediction and Marketplace Platform

Project Proposal for UCS503P

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1 Higher-order goal

...secondary framing

This project aims to improve real estate market transparency by providing predictive pricing and seamless buyer-seller communication. While the topic covers various property types like commercial and entertainment zones, the immediate engineering objective is to translate efficient property transactions into a measurable, deployable software solution.

2 Time-to-value

...primary heuristic

- We will deliver meaningful increments quickly by prototyping the core user workflow and validating it with users within a short iteration window.
- We will prioritize fast feedback over perfection by using weekly iterations (and targeting CI-backed automation early) to reduce release friction.
- Success is defined by what can be delivered and measured in the first iteration, then improved based on user feedback.

3 Problem Statement

Buyers and sellers face significant delays and uncertainty regarding property valuations and transaction coordination. Common pain points include:

- **Fragmentation:** complaints and communications are often made via multiple channels (calls, social media, paper forms), causing lost context.
- **Low transparency:** buyers cannot easily verify if a property is priced fairly based on current market trends.
- **Slow coordination:** buyers and sellers lack a unified, trackable workflow to negotiate and schedule site visits.

Why this matters now (contextual relevance): In a fast-moving real estate market, improving the communication and pricing process can noticeably reduce transaction delays and missed opportunities.

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4 Proposed Solution

... *Software Proposition*

4.1 Overview

Build a **web-based** real estate marketplace system with the following components:

- **Property Listing Portal:** Users submit a property with photos, area category (e.g., commercial, entertainment), and locality details.
- **Machine Learning Predictor:** Analyzes property data and outputs an estimated future price.
- **Buyer Barometer:** Visual indicators using eye-catching images to highlight properties selling at favorable rates.
- **Integrated Communication:** A WhatsApp-like chat for buyers and sellers to negotiate and generate site appointments.
- **Payment Integration:** Secure checkout via payment apps to confirm transactions or booking fees.

4.2 Core workflow

1. Seller lists a property and the system immediately generates a price prediction.
2. System validates completeness and suggests possible duplicates (lightweight similarity on text/category and approximate location).
3. Buyer discovers the property, guided by the visual Buyer Barometer.
4. Buyer and seller negotiate and schedule a site visit via the built-in chat.
5. Buyer confirms the agreement or appointment fee using the integrated payment app.

4.3 Operational constraints for an educational setting

- Keep the solution usable on low-to-mid range devices via responsive web design.
- Focus on workflow, automation, and system correctness over building complex deep-learning models.
- Design for deployability using common web hosting and managed databases.

5 Solution Approach

... *Engineering focus*

This is an engineering course project, so we focus on scalable administrative and workflow aspects rather than novel algorithm research.

5.1 Machine Learning pricing assistance

... *predictive modeling*

Price prediction will be regression-based. We will implement a Multiple Linear Regression model where the predicted price ($\hat{y}$) is calculated as:

$$\hat{y} = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \dots + \beta_n x_n + \epsilon$$

To evaluate the accuracy of these predictions against actual closing prices, we will measure the Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

5.2 Web architecture

... web-based solution

A typical 3-tier web architecture:

- **Frontend:** responsive UI for buyers and sellers (listings, chat interfaces, barometer dashboards).
- **Backend API:** authentication, listing submission, ML prediction endpoints, WebSocket chat handling, and payment gateway logic.
- **Database:** store users, property data, chat history, attachments metadata.

5.3 CI/CD and fast delivery enabler

CI/CD will be used to:

- Automatically build and run tests on every change.
- Produce repeatable deployment artifacts to staging.
- Enable safe and frequent releases of incremental features.

6 Evaluation Criterion

... measurable and attributable

We define success using measurable metrics tied to the core workflow.

6.1 Primary evaluation metric

Time-to-appointment: time between a buyer's first message and a confirmed site visit in the system.

- **Target:** median time ≤ 2 hours during pilot window.
- **Attribution:** measured from database timestamps (first message vs. appointment confirmation by seller).

6.2 Secondary evaluation metrics

- **Prediction Accuracy:** keeping the ML model's MSE within a 10% variance of actual local market prices.
- **Engagement Rate:** percentage of properties utilizing the chat negotiation tool.
- **Reliability:** availability of staging/production for login and submission during business hours (e.g., $\geq 99\%$ uptime in pilot).

6.3 Pilot validation plan

... high-level

- Recruit 10–20 pilot users (citizens) and 3–5 staff roles (department simulation if real staff unavailable).
- Compare metrics before/after within the iteration window by logging baseline estimates via short interviews or existing process observations.

7 Scalability

... with foresight

The proposition should scale in both theoretical and practical senses.

1. **Scalable (theoretically):** The system should support growth in number of properties and concurrent dashboard usage by:
 - decoupling components (frontend/backend),
 - enabling pagination and indexing for common queries,
 - using stateless API design.
2. **Operationally deployable (practically):** The system should run on common web hosting:
 - managed database,
 - containerized or platform-hosted deployment,
 - CI/CD pipeline for staging/production.

8 Engine availability heuristic

A mature set of “engines” (tooling/services) is available to implement this as a web-based project:

- Web framework (e.g., Flask/Django or Node.js) for REST APIs and WebSocket communication.
- Managed relational or document database (e.g., PostgreSQL or MongoDB).
- ML libraries (e.g., scikit-learn) for the linear regression model.
- Payment gateway API (e.g., Stripe, Razorpay) for transaction confirmations.
- CI runner and staging deployment target.

9 Project scope and deliverables

... iteration-friendly

9.1 Initial deliverable

... first iteration

- Property submission form + basic Linear Regression price output.
- Buyer Barometer UI implementation on the listing page.
- Basic logging and timestamps for Time-to-appointment measurement.
- CI: build + backend unit tests + linting.
- CD to staging with a single command or pipeline job.

9.2 Subsequent deliverables

- Real-time WhatsApp-like chat layer via WebSockets.
- Payment app integration for appointment booking fees.
- Admin dashboard for market metrics and active negotiations.

10 Risks and mitigations

- **Data privacy / consent:** minimize collected data; securely manage payment tokens.
- **Model Inaccuracy:** start with a baseline Linear Regression model and update weights as more property data is collected.
- **Evaluation measurement ambiguity:** instrument timestamps and define clear status transitions before development begins.
- **Operational issues in pilot:** use staging early; ensure CI/CD produces repeatable deployments.

11 Summary

This project proposes a deployable web platform to reduce transaction delays in the real estate market. It meets the course criteria by emphasizing time-to-value through rapid iterations, implementing CI/CD to support frequent safe releases, and using measurable evaluation criteria (especially time-to-appointment). It remains focused on engineering workflow, practical ML deployment, and scalability readiness rather than research-driven novelty.